

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Linear Graphs ($y = mx + c$)

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

LINEAR GRAPHS ($Y = MX + C$) · CH3 · L4



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The Map

This lesson collapses Linear Graphs ($y = mx + c$) into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Read gradient and intercept

TRIGGER *"gradient", "tangent", "rate of change"*

02 Find equation from two points

TRIGGER *"work out", "find", a missing value*

03 Use parallel/perpendicular gradients

TRIGGER *"gradient", "tangent", "rate of change"*

04 Sketch from equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 8 website questions, 8 checked solutions, 3 local PDFs read.

01 Read gradient and intercept

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5y + 2x = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2H Q18

The line is

$$5y + 2x = 7$$

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{5}{2}$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Find equation from two points

TRIGGER *"work out", "find", a missing value*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{0 - (-3)}{2 - 0} = \frac{3}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2023 P2H Q9

Solve equation

This is a reasoning step: write one clear sentence, then continue the calculation.

The gradient is

$$\frac{0 - (-3)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 3$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use parallel/perpendicular gradients

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5y + 2x = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2H Q18

The line is

$$5y + 2x = 7$$

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{5}{2}$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Sketch from equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"gradient", "tangent", "rate of change"	Read gradient and intercept
"work out", "find", a missing value	Find equation from two points
"gradient", "tangent", "rate of change"	Use parallel/perpendicular gradients
"work out", "show", "hence", a familiar exam phrase	Sketch from equation
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "find", a missing value	Find the gradient
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.